

REPORT · AUGUST 2026

minimax-pdf demo

A demonstration of the token-based design pipeline

Executive Summary

This document was generated by minimax-pdf — a skill for creating visually polished PDFs. Every design decision is rooted in the document type and content, not a generic template.

Key insight: design tokens flow from palette.py through every renderer, keeping cover and body visually consistent.

How It Works

The Token Pipeline

The palette.py script infers a color palette and typography pair from the document type. These tokens are written to tokens.json and consumed by every downstream script.

- 1. palette.py generates color tokens, font selection, and the cover pattern
- 2. cover.py renders the cover HTML using the selected pattern
- 3. render_cover.js uses Playwright to convert the HTML cover to PDF
- 4. render_body.py builds inner pages from content.json using ReportLab
- 5. merge.py combines cover + body and runs final QA checks

Cover Patterns

Pattern	Document type	Visual character
fullbleed	report, general	Deep background · dot-grid texture
split	proposal	Left dark panel · right dot-grid
typographic	resume, academic	Oversized display type · first-word accent
atmospheric	portfolio	Dark bg · radial glow · dot-grid
magazine	magazine	Cream bg · centered · hero image

darkroom	darkroom	Navy bg · centered · grayscale image
terminal	terminal	Near-black · grid lines · monospace
poster	poster	White · thick sidebar · oversized title

Data Visualisation

Performance Metrics (Chart)

Charts are rendered natively using matplotlib with a color palette derived from the document accent. No external chart services or image files required.

[Chart: install matplotlib to render — pip install matplotlib]

Market Share (Pie Chart)

[Chart: install matplotlib to render — pip install matplotlib]

Mathematics

Display math is rendered via matplotlib mathtext — no LaTeX binary installation required. Inline references use standard [N] notation in body text.

$$E = mc^2$$

$$\int_0^{\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Process Flow

Flowcharts are drawn directly using matplotlib patches — no Graphviz or external tools needed. Supported node shapes: rect, diamond, oval, parallelogram.

[Flowchart: install matplotlib to render — pip install matplotlib]

Code Example

PYTHON

```
# Design token pipeline
tokens = palette.build_tokens(
    title="Annual Report",
    doc_type="report",
    author="J. Smith",
    date="March 2026",
)
html = cover.render(tokens)
pdf = render_cover(html)
```

Design Principles

The aesthetic system is documented in design/design.md. The core rule: every design decision must be rooted in the document content and purpose. A color chosen because it fits the content will always outperform a color chosen because it seems safe.

Restraint over decoration

The page is done when there is nothing left to remove. Accent color appears on section rules only — not on headings, not on bullets. No card components, no drop shadows.

A PDF passes the quality bar when a designer would not be embarrassed to hand it to a client.

References

- [1] Bringhurst, R. (2004). *The Elements of Typographic Style* (3rd ed.). Hartley & Marks.
- [2] Cairo, A. (2016). *The Truthful Art: Data, Charts, and Maps for Communication*. New Riders.
- [3] Hochuli, J. & Kinross, R. (1996). *Designing Books: Practice and Theory*. Hyphen Press.